RevIN Experiment 1: Normalization Components

1 Theoretical overview and goal

Global standardization applies training-wide statistics. Reversible instance normalization instead uses the current look-back:

$$\tilde{x} = \alpha \frac{x - \mu_x}{\sigma_x} + \beta, \qquad \hat{y} = \sigma_x \frac{f_\theta(\tilde{x}) - \beta}{\alpha} + \mu_x.$$

Non-affine instance normalization fixes $(\alpha, \beta) = (1, 0)$. It removes window-specific offset and scale, which can reduce temporal and cross-user shift, but it also discards potentially predictive level information.

Experimental goal. Isolate the contribution of global normalization, reversible per-instance normalization, and RevIN's learnable affine parameters under the same data-space training loss.

2 Implementation details

The primary MSE-trained comparison is `none_mse`, `standard_mse`, `instance_mse`, and `revin_mse`. Standard statistics are fitted on training data only. Instance and RevIN statistics are computed from each look-back and inverted after forecasting; all reported MSEs are in original data space.

The full sweep uses ETTh1 (all seven variables), Electricity, Traffic, Solar, Weather, and Exchange Rate; six (L, H) settings; PatchTST; three seeds; batch size 256; learning rate 10^{-5}; and exactly 10,000 optimizer steps. The long-horizon setting is $(336, 720)$. Validation and logging run every 1,000 optimizer steps, while evaluation windows use a horizon-sized stride. Training windows are sampled at random over users and dates. Test mode uses Electricity/Solar, two settings, PatchTST, seeds 1–2, and 2,000 optimizer steps.

Implementation: `src/utils/models.py`
Launcher: `revin.slurm`

Run isolated `test` mode first. The `small` profile then reproduces the original Traffic/Electricity/Solar PatchTST study on four settings. The `full` profile shares its output root, reuses completed small seeds, and adds all datasets, two comparison settings, standard–nMSE, last-value, and arcsinh runs. The `ultra` profile adds DLinear. One table is produced per model and test split, with seed mean ± sample standard deviation and an explicit row multiplier $\times 10^m$. JSON summaries also retain the variance of per-point losses.

3 Interpretation

The none–standard contrast measures global conditioning. Standard–instance measures the value of sample-specific alignment. Instance–RevIN isolates the affine parameters; it should not be described as the effect of instance normalization itself.